Karen Xiao

Curriculum Vitae

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EDUCATION

Wellesley College, *Wellesley, MA* 2023 - 2026
B.A. in Computer Science and Mathematics, *Summa Cum Laude*
Departmental Honors in Computer Science
Thesis Advisor: Brian Brubach

University of Maryland, College Park, *College Park, MD* 2022 - 2023
Honors College, Gemstone Research Program

RESEARCH EXPERIENCE

Undergraduate Honors Thesis, *Wellesley College* Sep 2025 - May 2026
Thesis Title: Quantifying Diversity of Educational Evaluation Using Relative Student Performance
Advised by Professor Brian Brubach

- Completed a senior honors thesis that proposes methods to measure the diversity of evaluation in classes by treating different types of assignments as alternative ways of ranking students

NSF REU Intern, *CMU Human-Computer Interaction Institute* May 2025 - Aug 2025
Project: Exploring the Impact of Data-Driven Optimization of Tutoring Systems on Student Learning
Advised by Conrad Borchers and Professor Vincent Aleven

- Applied statistical methods to analyze the impact of data-driven redesign on intelligent tutoring systems
- Identified new insights into what constitutes a productive level of difficulty for data-optimized practice

Research Assistant, *Institute for Mathematics and Democracy* Dec 2024 - Sep 2025
Project: Empirical Investigation of Ranked Voting Methods
Advised by Professors David McCune, Ismar Volić, and Andy Schultz

- Simulated ranked choice election methods on datasets containing ballots for thousands of real elections
- Conducted empirical comparison of various ranked choice voting methods using Python and VoteKit
- This research is believed to be the largest study of ranked choice elections ever performed

Undergraduate Student Researcher, *Wellesley College CrowdNets Lab* Aug 2024 - May 2025
Project: A Graph-Based Spatio-Temporal POI Ranking Measure for Pickup and Delivery Platforms
Advised by Professor Christine Bassem

- Defined and evaluated a graph-based spatio-temporal POI ranking model that outperforms baseline models
- Ran simulations and designed dynamic visualizations on real-world Uber data to evaluate performance
- Won 2nd place in the Undergraduate Student Research Competition at ACM SIGSPATIAL 2025

Undergraduate Student Researcher, *MIT Urban Metabolism Group* Feb 2024 - May 2024
Project: Urban Climate Policy Mapping with AI
Advised by Norhan Bayomi

- Collected and analyzed data regarding climate risks and policies in various metropolitan clusters
- Converted a policy portfolio into a quantifiable matrix for ML-driven insights

PUBLICATIONS

Peer-Reviewed Conference Proceedings (Short Papers)

1. Liu, Q., Borchers, C., Xia, M., **Xiao, K.**, Carvalho, P., Koedinger, K., & Aleven, V. (2026). Evaluating a Process for the Datadriven Redesign of Educational Technology: Improved Learning Processes. In *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED '26)*. https://doi.org/10.1007/978-3-032-29760-0_12
2. **Xiao, K.**, Irisumi, M., & Bassem C. (2025). A Graph-Based Spatio-Temporal POI Ranking Measure for Pickup and Delivery Platforms. In *Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25)*. https://doi.org/10.1145/3748636.3766535 [**SRC 2nd Place Winner**]

Peer-Reviewed Conference Proceedings (Highlighted Posters)

1. Brubach, B., Xiao, K. (Upcoming Nov 2026). Quantifying Diversity of Educational Evaluation Using Relative Student Performance. In *Proceedings of the 6th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (ACM EAAMO '26)*.

TEACHING EXPERIENCE

Teaching Assistant

Wellesley College

CS 231 Design & Analysis of Algorithms — Spring 2024 - Spring 2026
STAT 218 Introductory Statistics & Data Analysis — Spring 2026

HONORS AND AWARDS

Awards

Excellence in Teaching Award in Computer Science, *Wellesley College* — May 2026
NSF GRFP Honorable Mention, *National Science Foundation* — Apr 2026
Undergraduate SRC 2nd Place Winner, *ACM SIGSPATIAL '26* — Nov 2025

Conference Funding

SIGSPATIAL Travel Award (SIGSPATIAL '26) - $1,000, *National Science Foundation* — 2025
Science Center Student Conference Travel Grant - $300, *Wellesley College* — 2025

WORK EXPERIENCE

Financial Data Analyst Intern, *Quantbot Technologies* — Summer 2023, 2024, 2026
Team: Data Trading Lab

LEADERSHIP EXPERIENCE

Co-Captain, *Wellesley Club Squash Team* — 2025-2026
Networking Chair, *Wellesley Computer Science Club* — 2024-2026